

MACSS Capstone Project

With the California Research Bureau at the California State Library

Sanjana Karthick
Shun Moriguchi

May 1st, 2026

A dark blue diagonal gradient bar that starts from the bottom left corner and extends towards the top right corner, covering the lower half of the slide.

Project overview

- Parsed **CA budget bills** for budget items
- Traced **spending** from Open Fi\$Cal to budget items
 - To see **how much** money budgeted is being spent.
- Merged **vendor locations** for each spending item
 - To see **where** money is being spent
- Focusing on **2023** as proof of concept

Our goal

Building a prototype pipeline can serve as proof of concept for CRB to request funding

With funding, CRB could outsource development of a comprehensive, queryable database of budget and spending data across time.

- Eliminates repetitive, one-off research
- Enables faster, more efficient budget and spending analysis

Legislators want to know if their Constituencies are receiving and spending money appropriately

Data Description & Preparation

- **2023 Budget Acts**
 - Text data
 - There is one original bill and three “trailer bills”
 - Scraping and parsing them for
 - Item number
 - program / object code
 - Amount
 - Description

0250-001-0001—For support of Judicial Branch		
.....		642,871,000
Schedule:		
(1)	0130-Supreme Court	53,140,000
(2)	0135-Courts of Appeal	271,988,000
(3)	0140-Judicial Council	311,639,000
(4)	0155-Habeas Corpus Resource Center	18,597,000
(5)	Reimbursements to 0140- Judicial Council	–12,293,000
(6)	Reimbursements to 0135- Courts of Appeal	–200,000

Data Description & Preparation

- **Open Fi\$Cal**
 - Ledger book for state transactions
 - Vendor name
 - Accounting date
 - Spending amount
 - Merged vendor locations
 - Location data is based on the company's location, which may differ from the location of the service being rendered.
- **Crosswalk tables**
 - To connect budget bill data to actual spending data
 - Created “Budget ID” based on budget codes to identify individual budget items
 - Successfully traced over 99% of spending transactions to budget items

EDA: Fi\$Cal Location Data

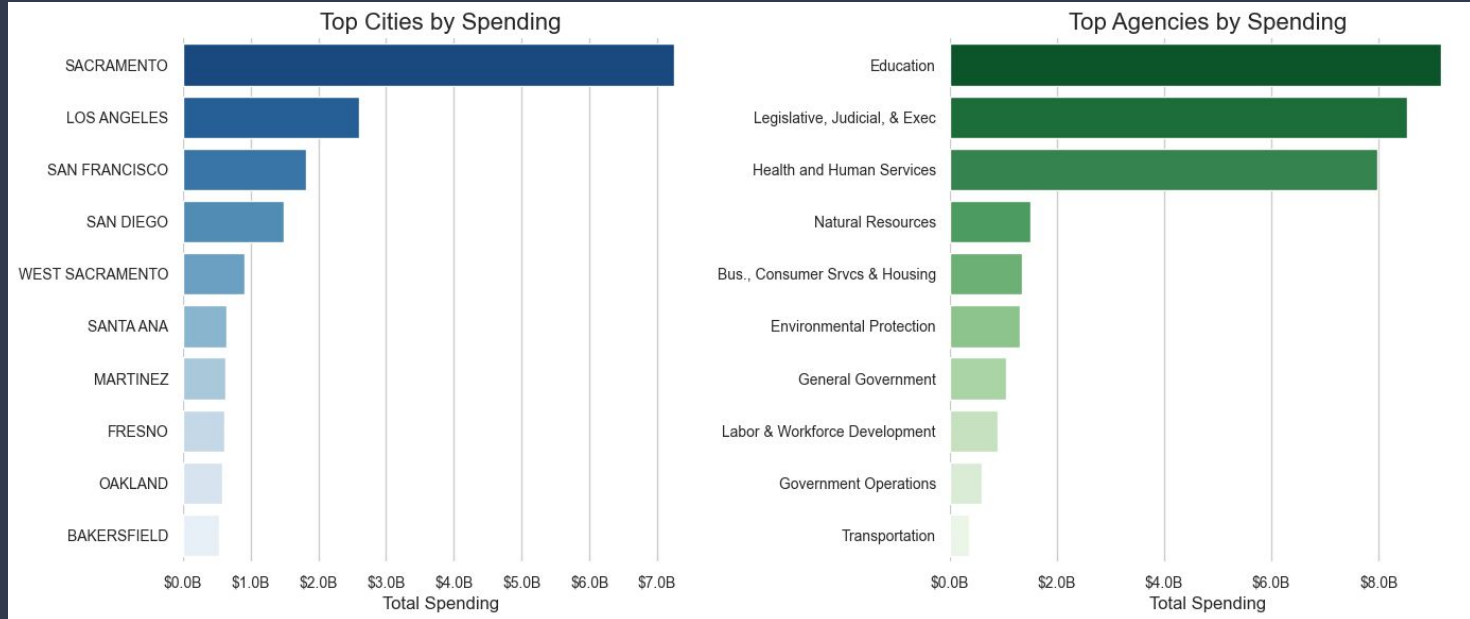
Total rows: 2,380,960

Missing addresses: 349,168 (14.67%)

Total spend: \$32,920,835,663

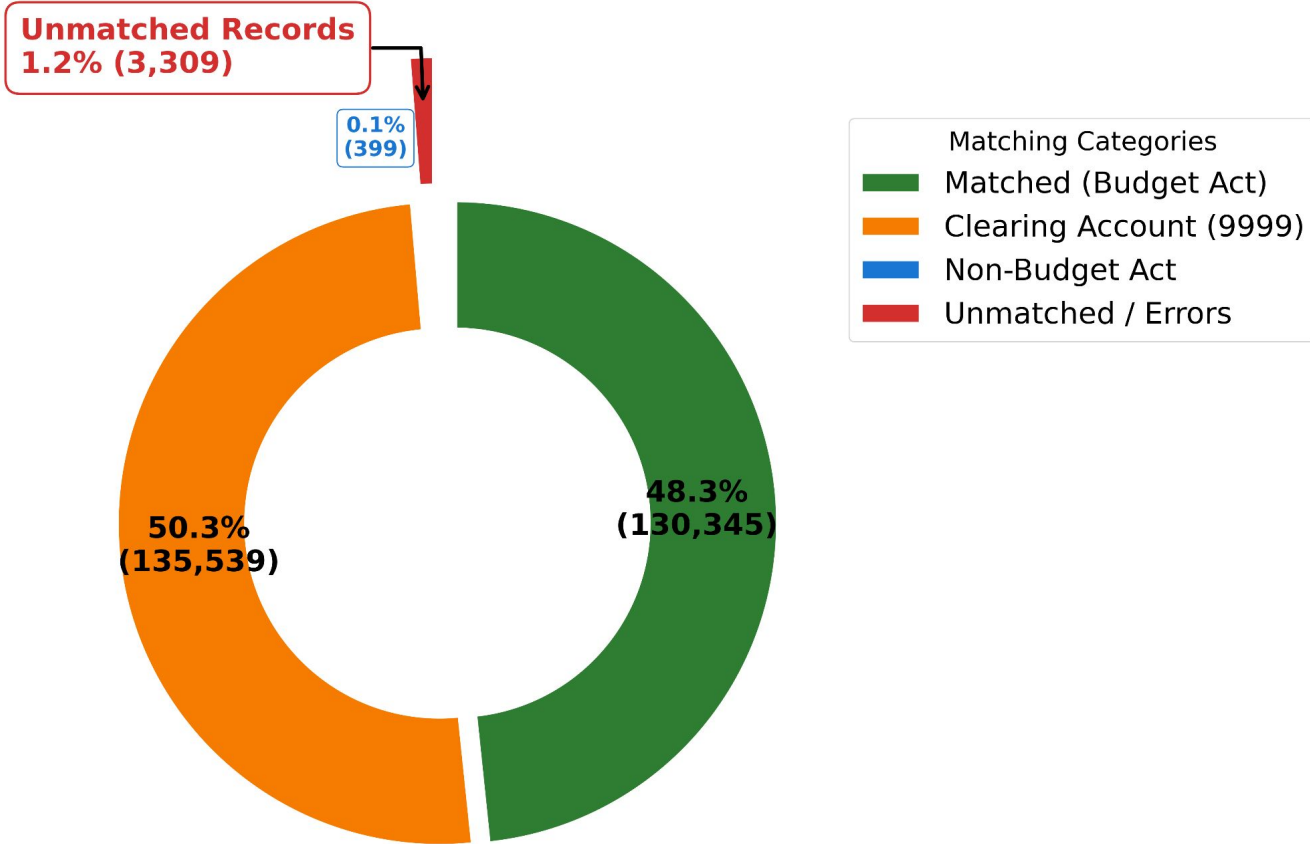
Missing spend addresses: \$1,350,362,126 (4.10%)

EDA: Fi\$Cal Location Data



May 2024 (FY2023) Matching Status Analysis

Total: 266,283 Records



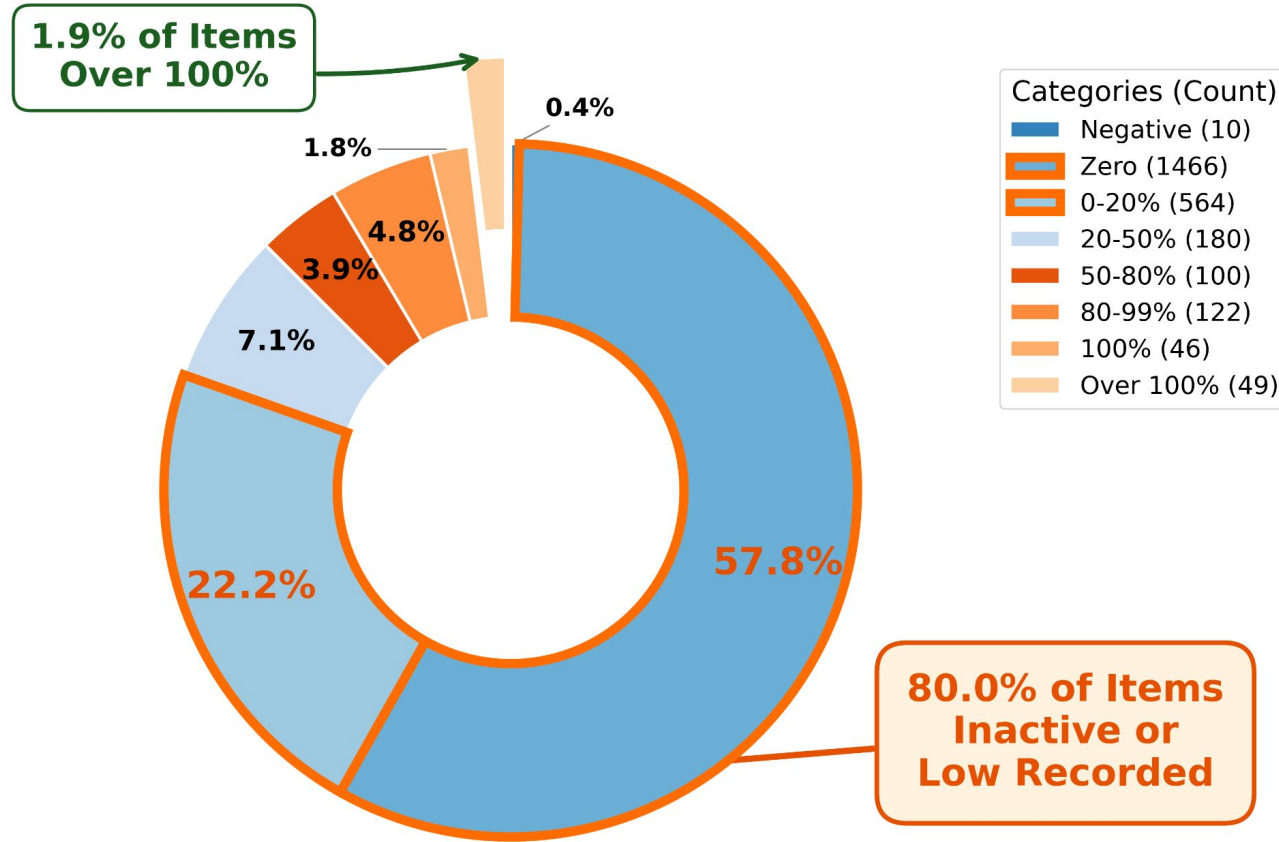
Our findings

Comprehensive Expenditure Analysis (30-Month Period)

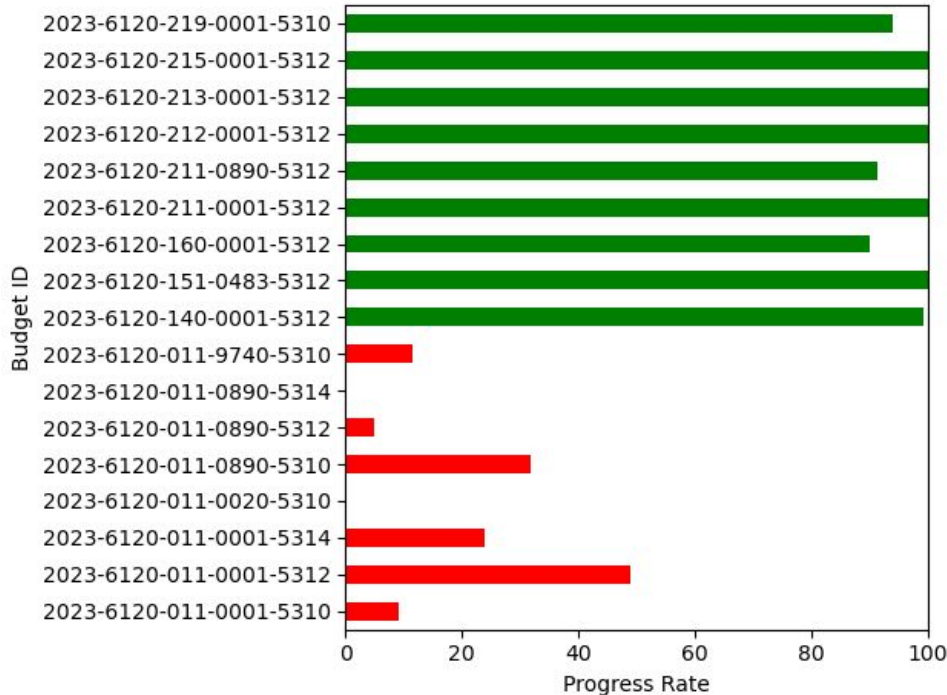
Analyzed 30 months of expenditure data through December 2025.

- The overall spending rate stands at 8.25%.
- 60% of budget line items show 0% utilization.
- 20% of items show a utilization rate of less than 20%.
- Certain programs exceed 100% utilization.

Fiscal Year 2023: Budget Utilization Distribution



Analysis of budget on California State Library



“Local assistance”

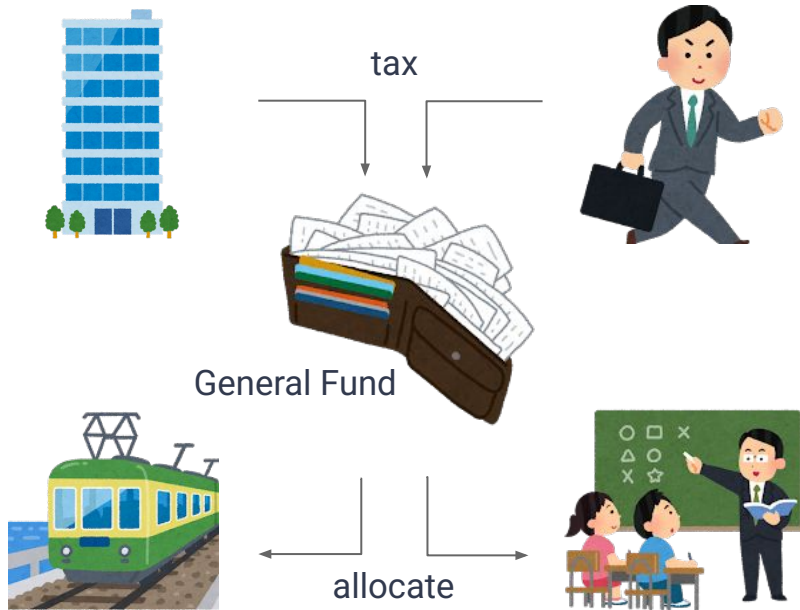
Pass-through funds for counties and cities supporting local literacy, infrastructure, and regional library services.

“Support”

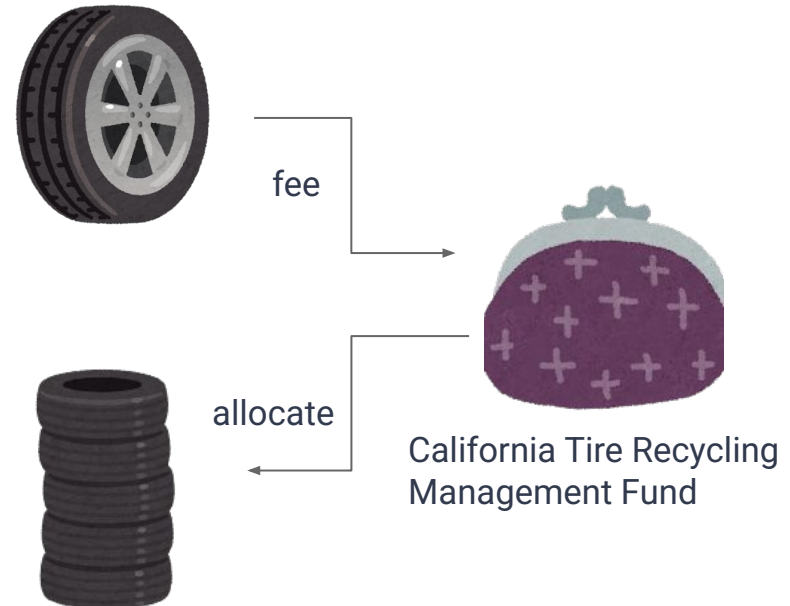
Internal funds for legislative research, archive preservation, and maintaining statewide information technology infrastructures.

Progress Rate Exceeding 100%: California Tire Recycling Management

General Business Operations



California Tire Recycling Management



Takeaways

- 30% vendor locations missing
- The spending rate is low.
 - Current theories:
 - Majority of spending is intergovernmental transfers
 - Employee payroll
 - Spending is still in progress
- California tracks budgets by agency, but does not track spending/ spending locations comprehensively.
- Improving statewide spending and location tracking should be a growing priority.

Future work

- Set up schedule/system to update vendor location data periodically (work with fiscal procurement team)
- Collect more comprehensive spending data - address the gaps in Open Fi\$Cal
- Expanding the database to all years
- Build an interface tool

Thank you!

Any Questions?

A dark blue diagonal gradient bar that starts from the bottom left and extends towards the top right, covering the lower half of the slide.